

A Design Study on Visual Analytics for EEG Artefact Removal

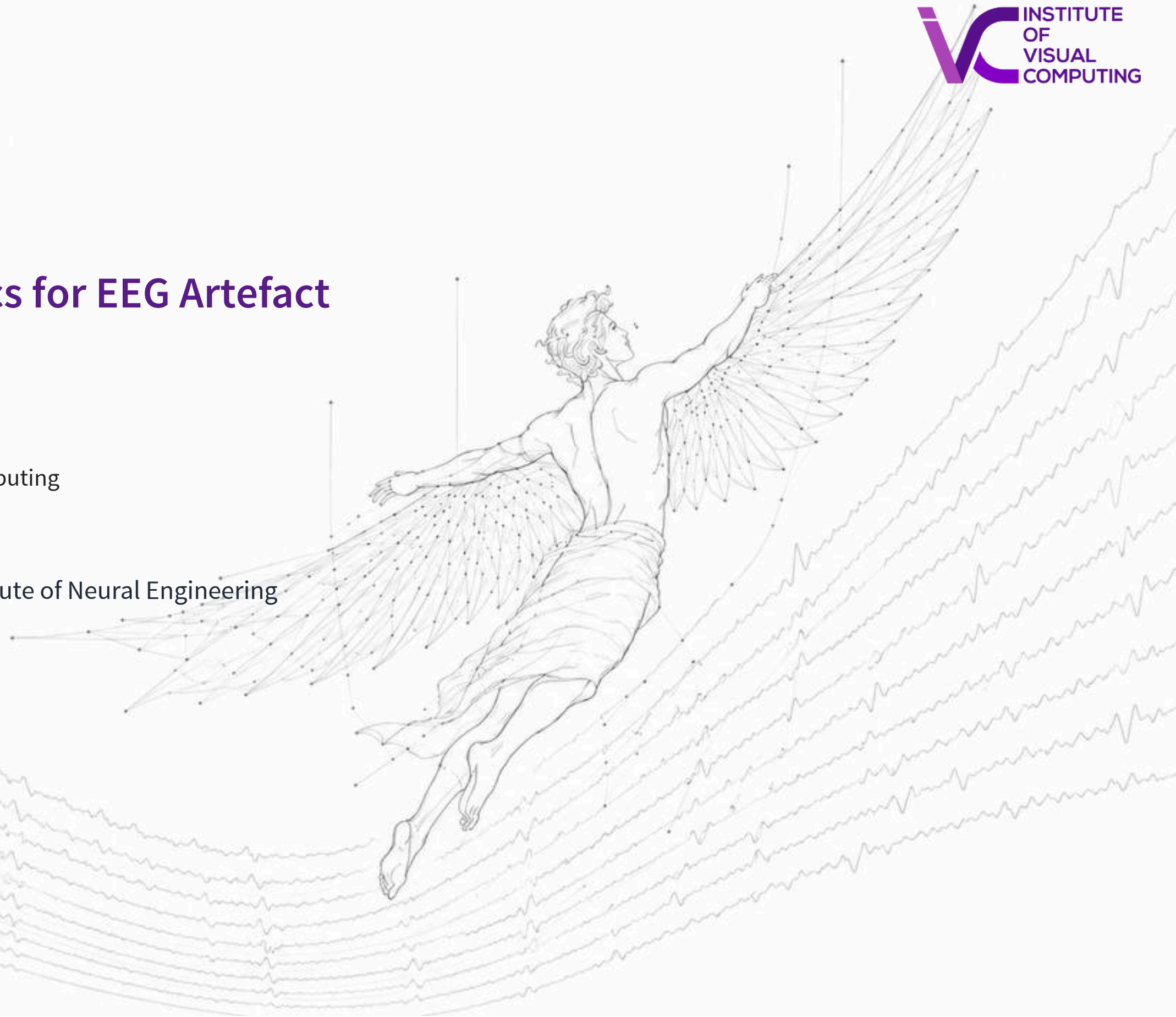
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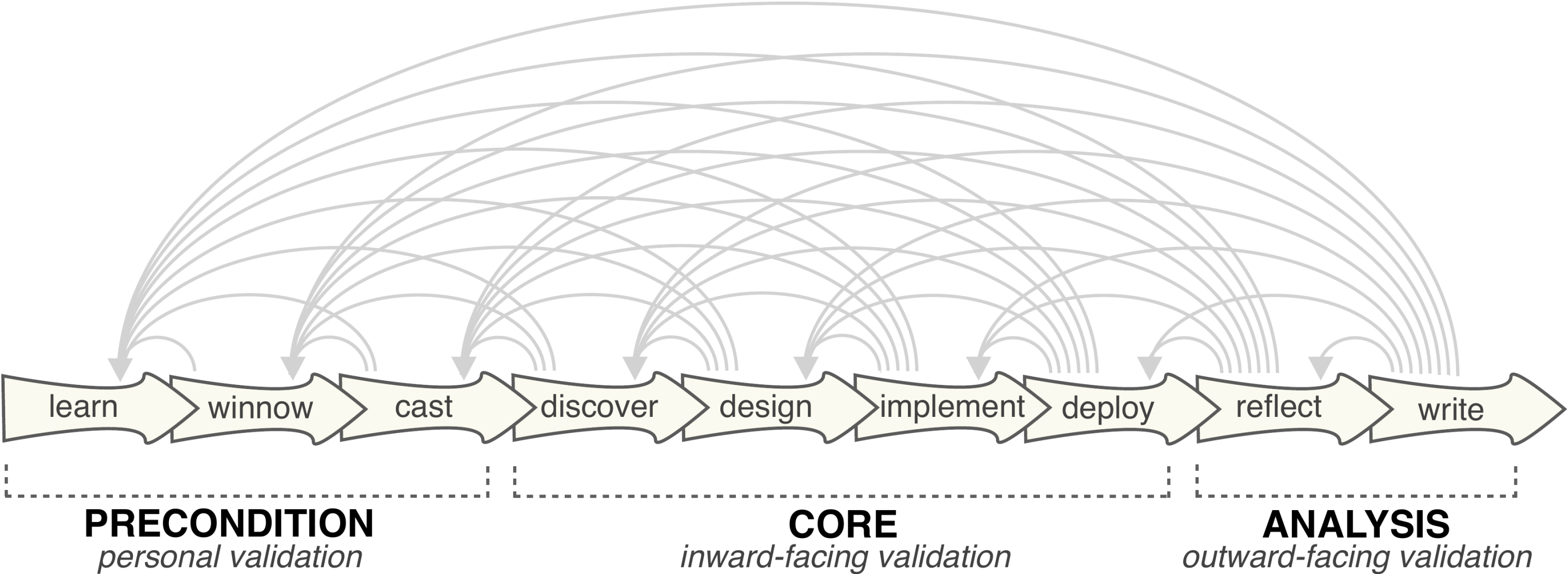
Supervisors: Julian Rakuschek, Tobias Schreck

Consultants: Michael Leitner, Selina Wriessnegger · Institute of Neural Engineering

Title image generated with ChatGPT, 2026
25 June 2026

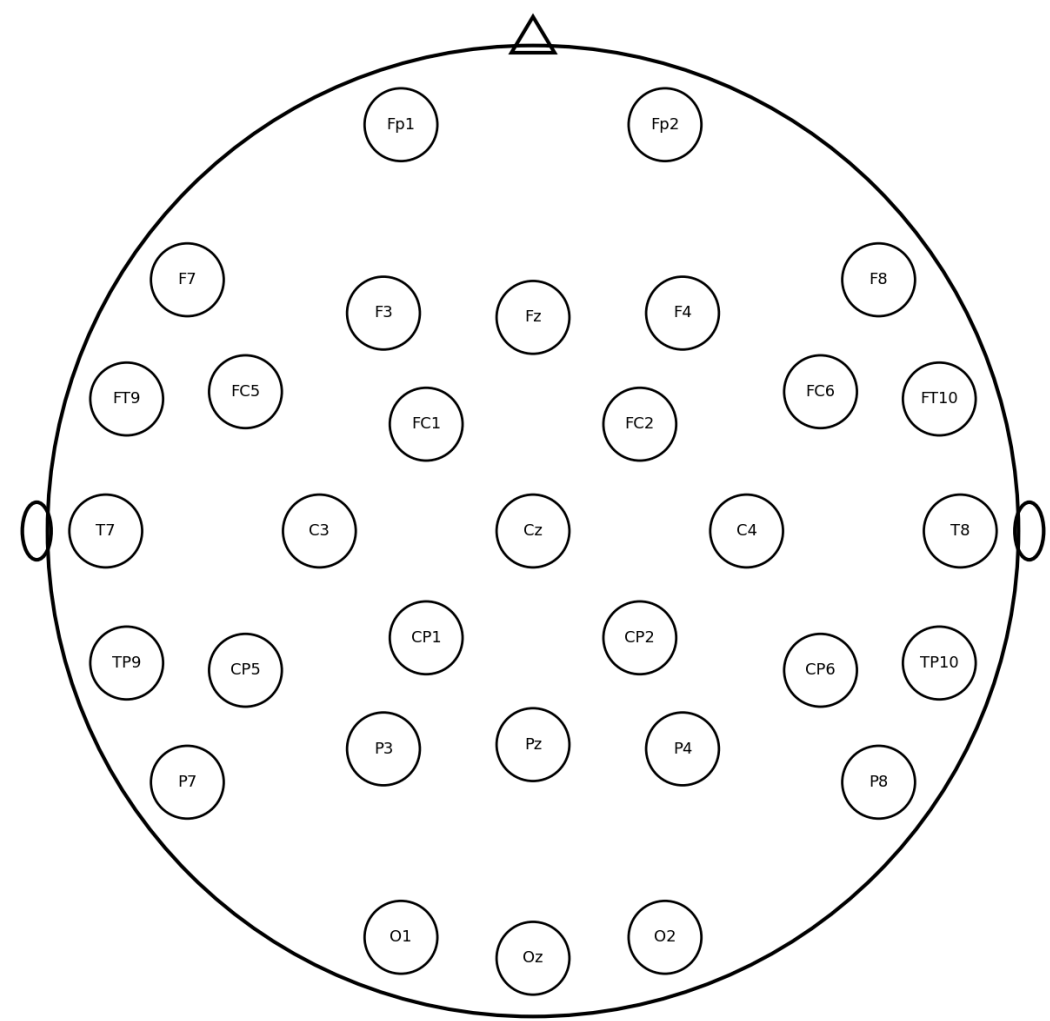


Design Study Methodology (Sedlmair et al.)

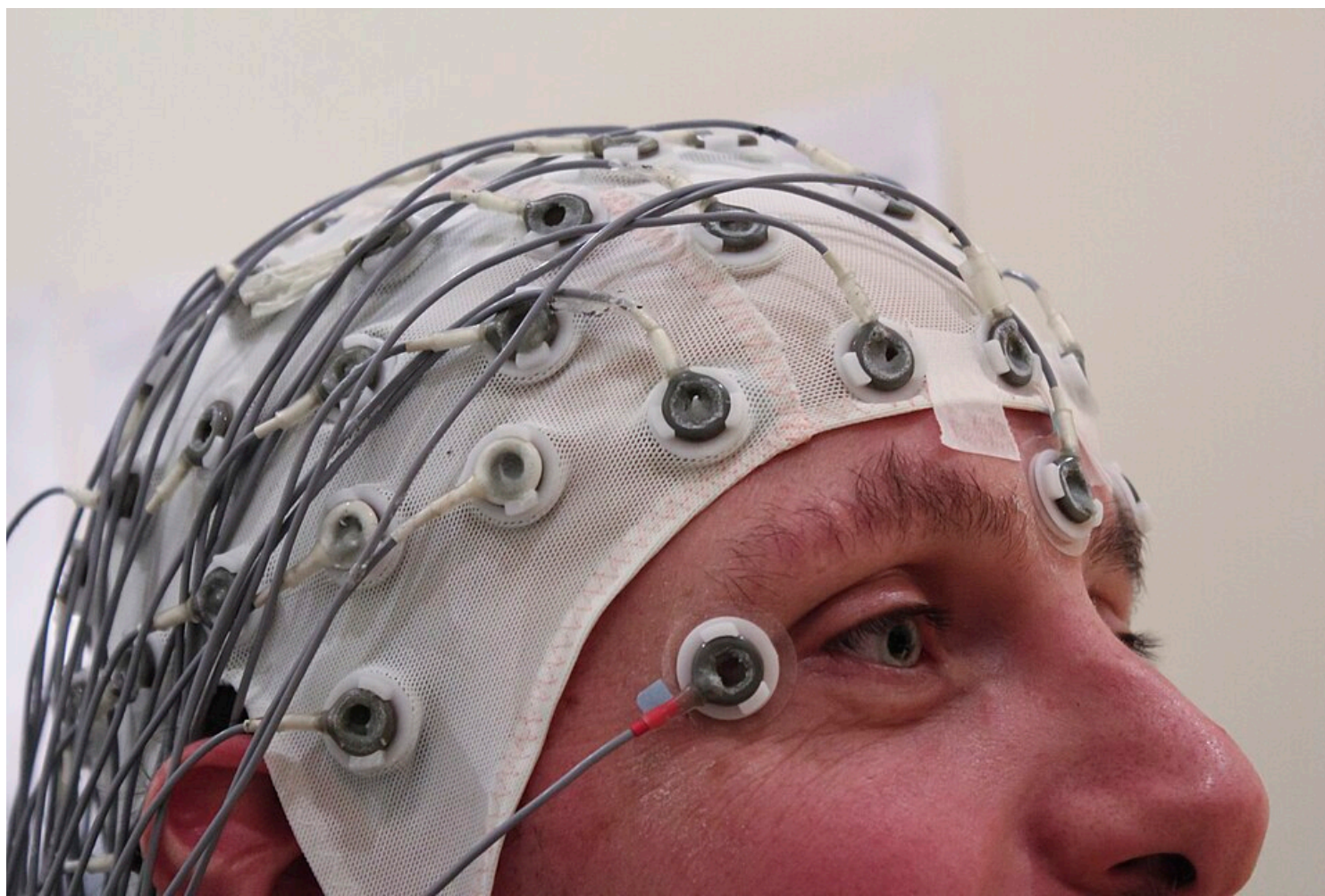


Sedlmair et al., *Design Study Methodology*, IEEE TVCG 18(12):2431–2440, 2012. © IEEE

Electroencephalography (EEG)

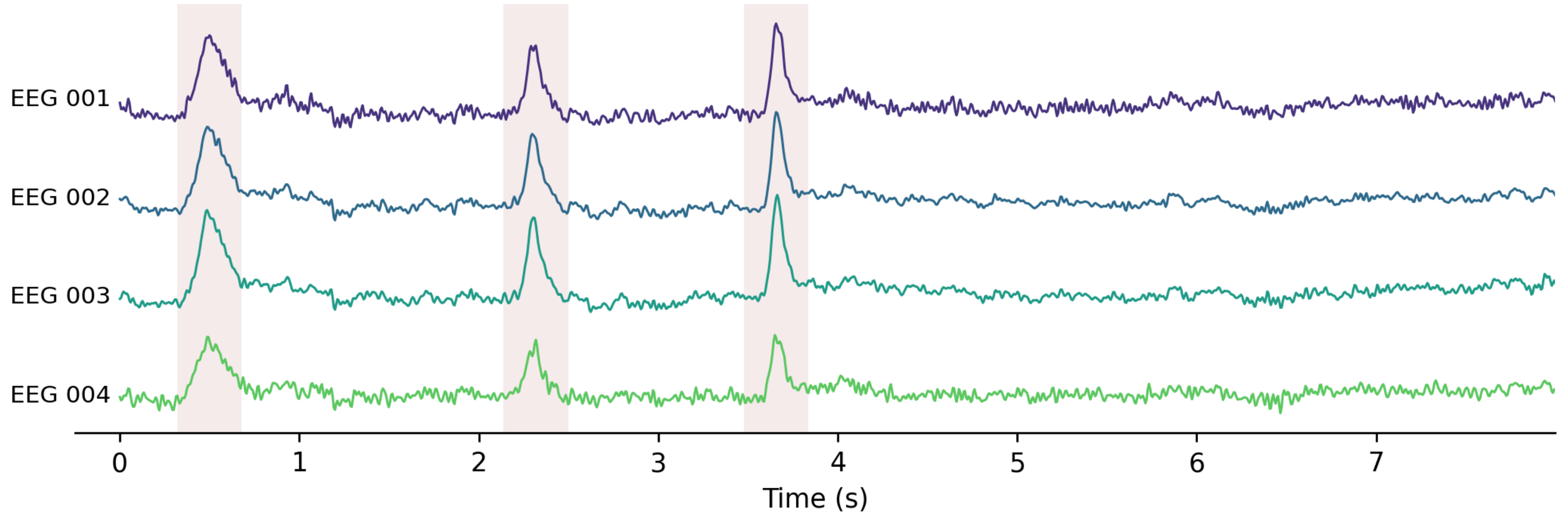


32-channel 10-10 electrode layout

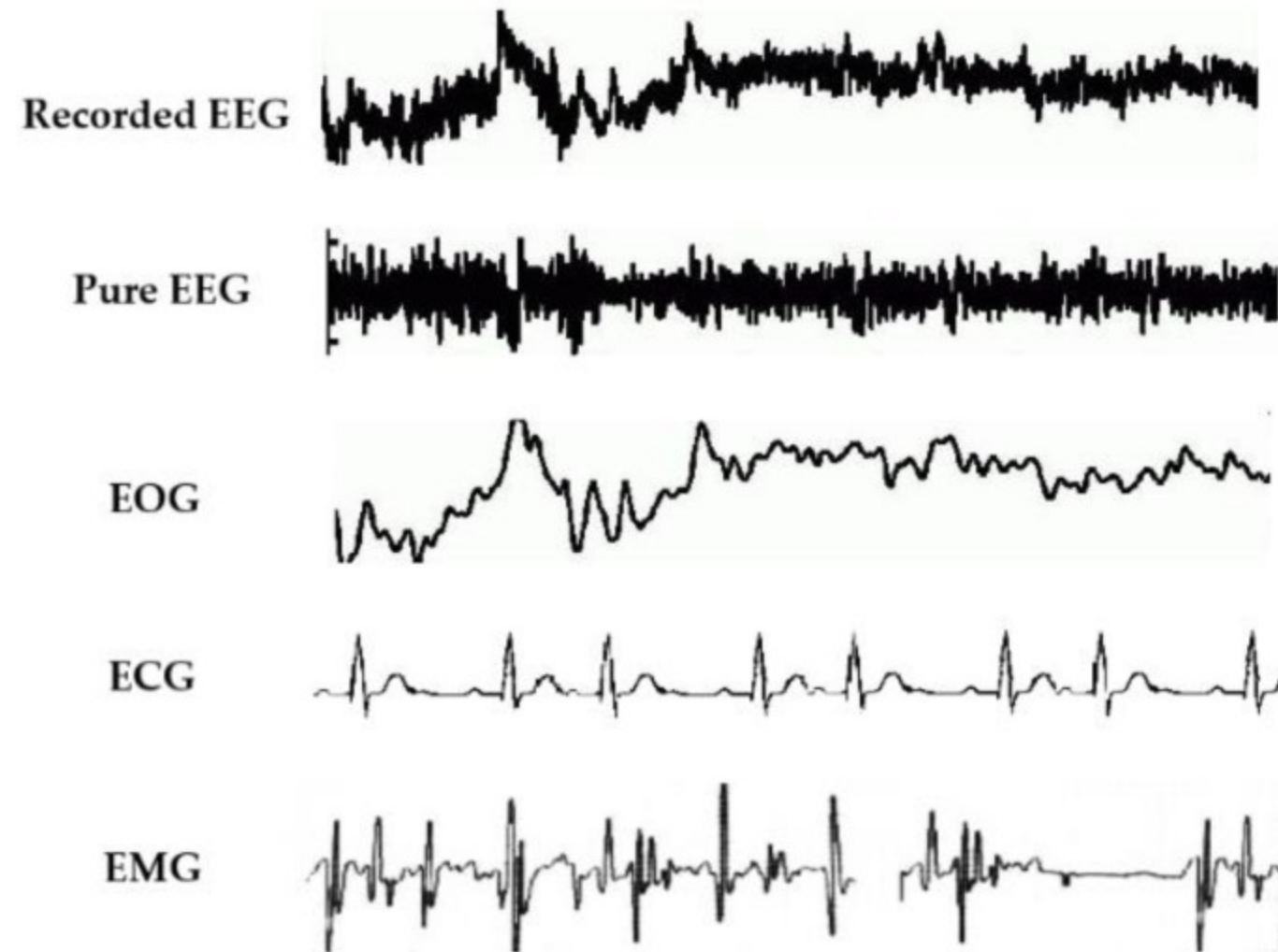


"EEG Recording Cap" by Chris Hope · CC BY 2.0

EEG Artefacts



Artefact Types



Jiang et al., *Removal of artifacts from EEG signals: a review*, Sensors 19(5):987, 2019. CC BY 4.0

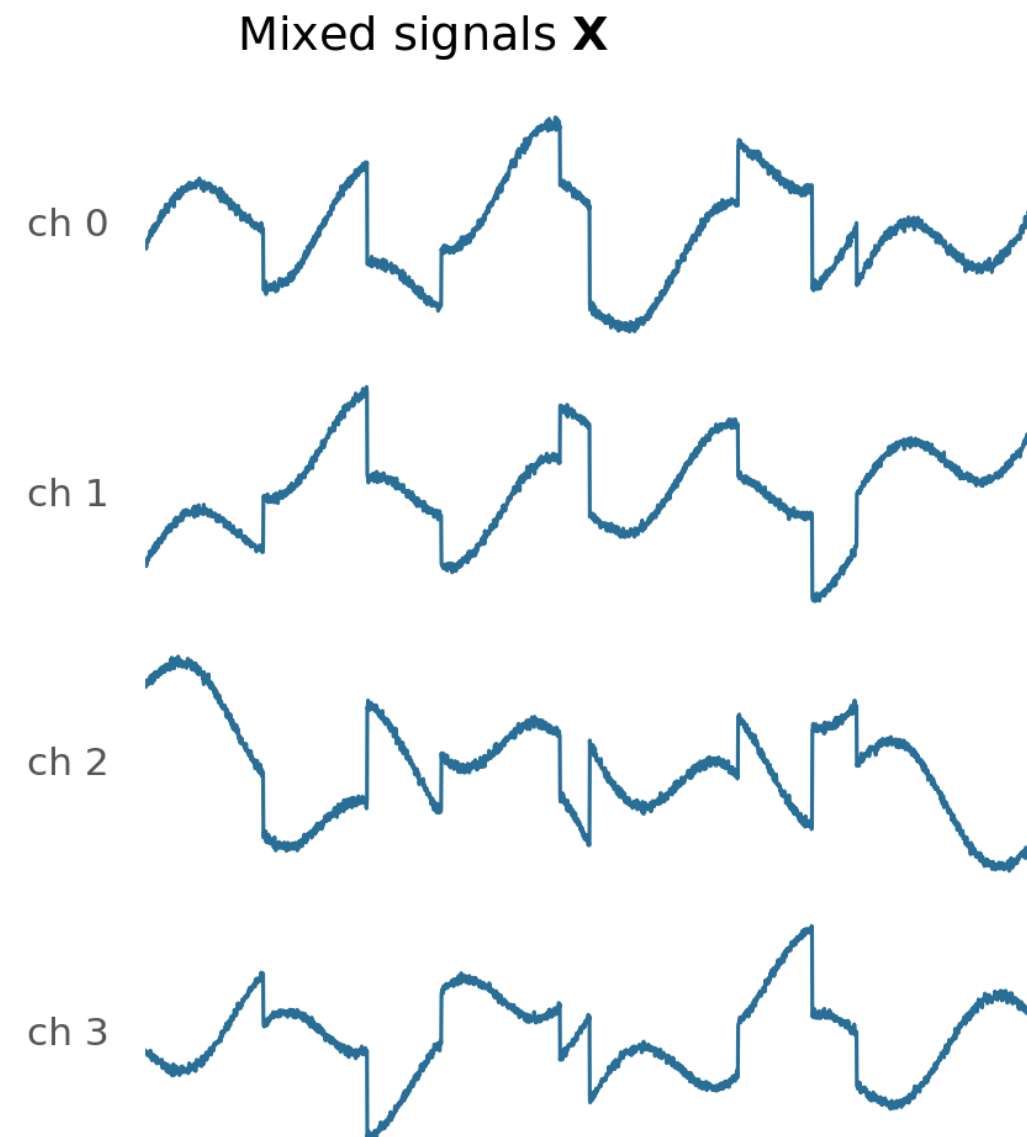
Physiological

- Ocular
- Cardiac
- Muscle

Technical

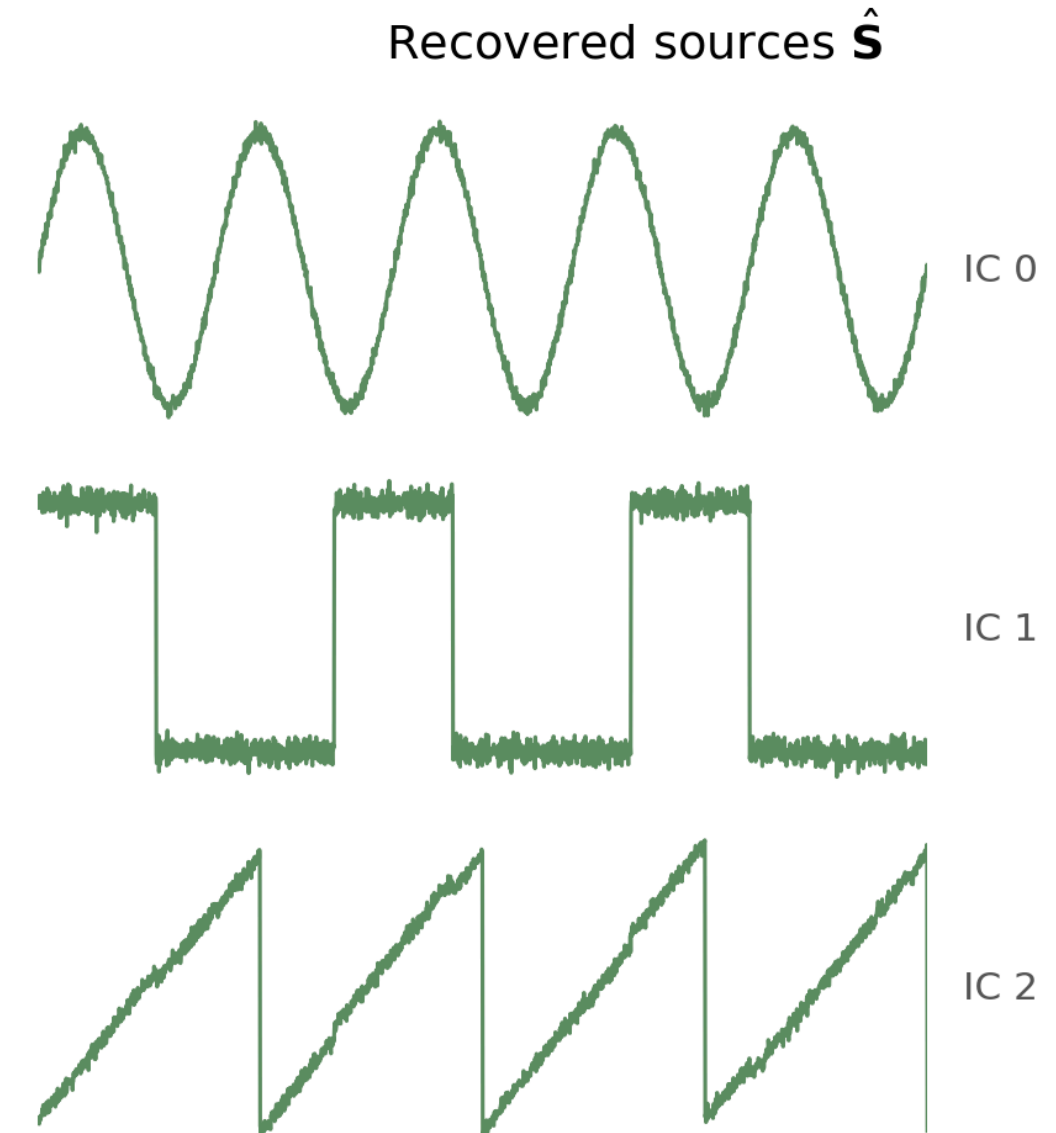
- Line noise
- Electrode / channel noise
- Movement
- Environmental

Independent Component Analysis



FastICA

$$\hat{\mathbf{S}} = \mathbf{W} \mathbf{X}$$



$$\mathbf{X} = \mathbf{A} \cdot \mathbf{S} \quad \rightarrow \quad \hat{\mathbf{S}} = \mathbf{W} \cdot \mathbf{X}$$

Manual Component Classification

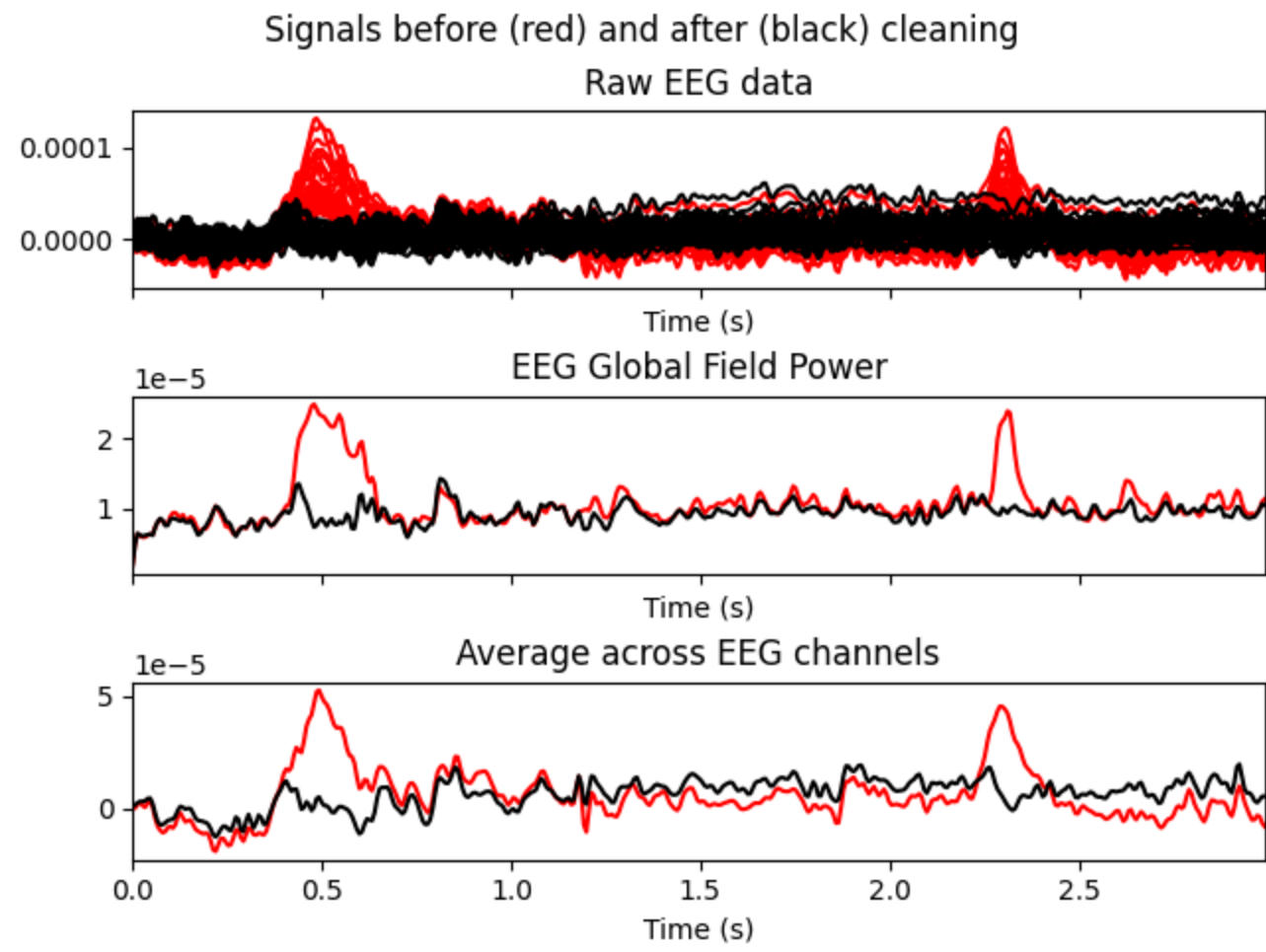


Figure: MNE-Python, `ica.plot_overlay` (sample dataset), BSD-3-Clause. Gramfort et al., *MEG and EEG data analysis with MNE-Python*, Front. Neurosci. 7:267, 2013.

Existing Tools

Capability	MNE-Py	FieldTrip	EEGLAB	ICLabel	MARA	ADJUST
<i>Computational primitives</i>						
ICA decomposition	●	●	●	—	—	—
Power spectrum	●	●	●	—	—	—
Algorithmic decision	● ^a	○	○ ^b	●	●	●
Confidence score	● ^a	○	○ ^b	●	●	○
Cross-subject matching	●	○	●	—	—	—
<i>Interactive workflow</i>						
Single-component view	●	●	●	—	—	●
Linked view updates	○	○	○	—	—	—
Cross-participant browser	○	○	●	—	—	—
Component clustering view	○	○	● ^b	—	—	—
Batch apply recommendations	○	○	○	—	—	—
Label export	●	●	●	●	●	●

● supported · ○ absent in core · ● partial. ^a per-artefact detectors, no unified classifier/calibrated probability. ^b via plugins.

Research Questions

RQ1

Which visualisation and interaction techniques support expert decision-making in ICA artefact classification?

RQ2

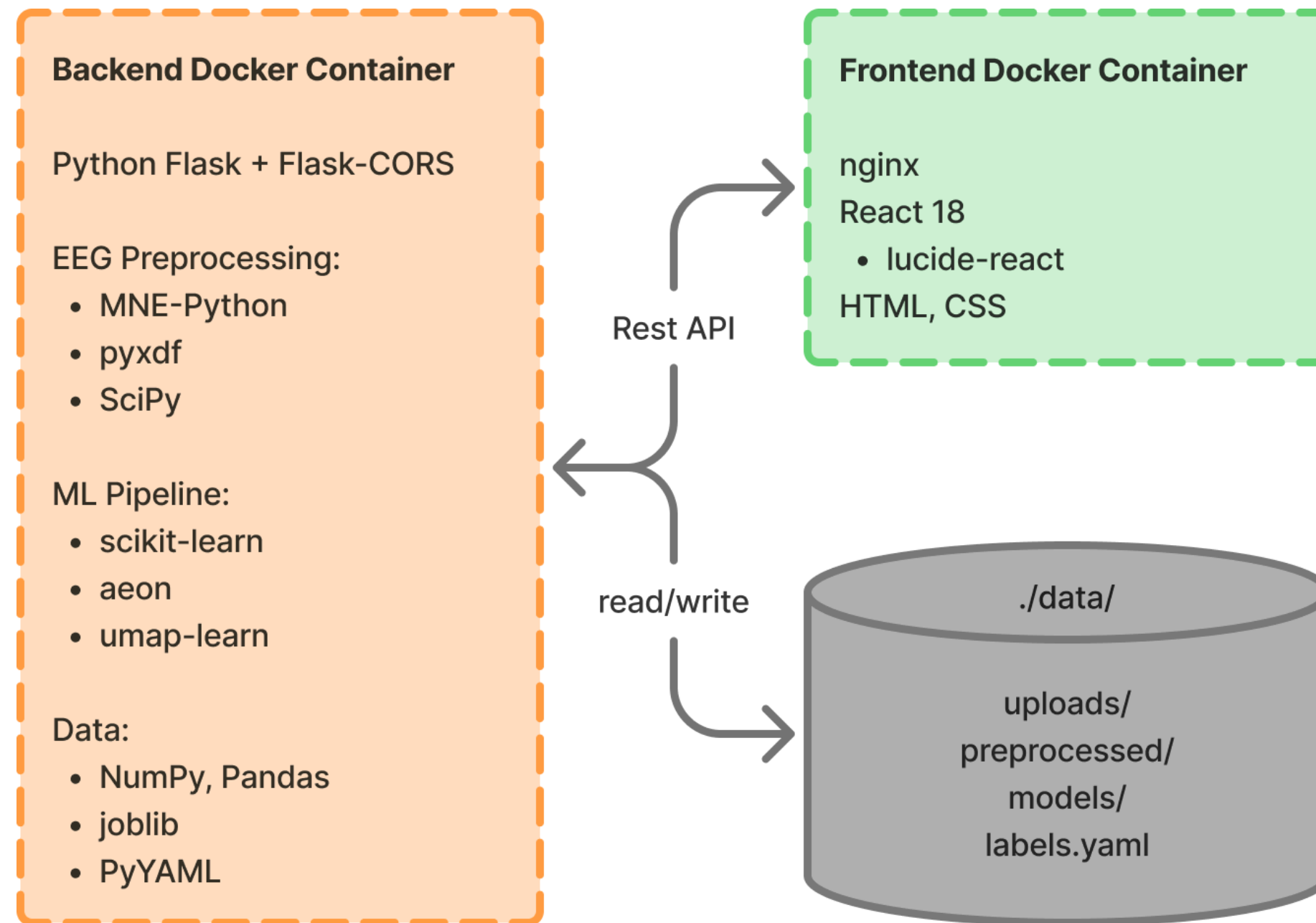
How can machine learning recommendations and interactive visualisation techniques be integrated to accelerate classification while preserving expert control?

Requirements

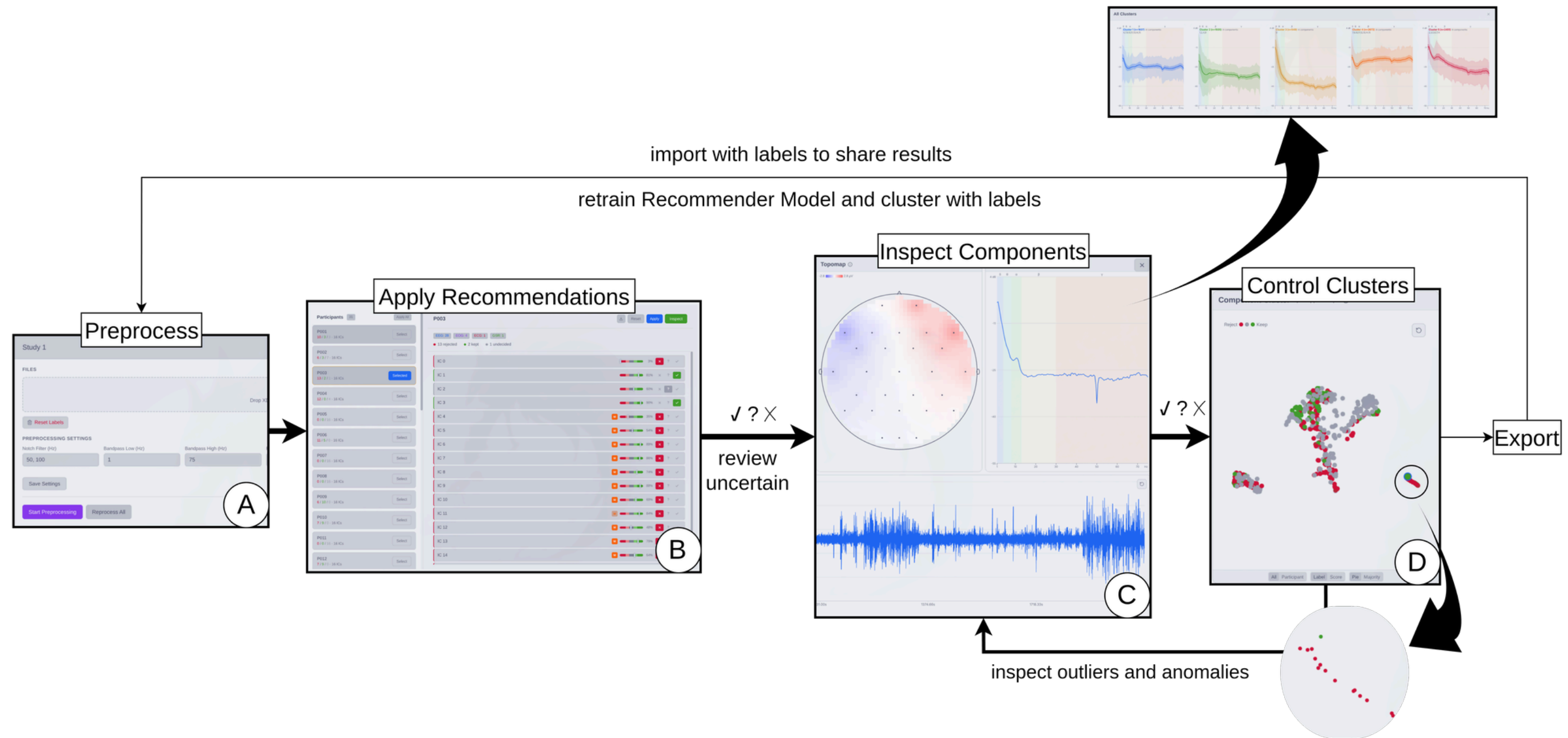
The system should support the user in...

- **R1** ...receiving visual recommendations on which components to keep or reject.
- **R2** ...gaining an overview of all components and supporting exploration, pattern recognition, and context for uncertain classification decisions.
- **R3** ...inspecting when and how artefacts appear in the recording over time.
- **R4** ...visually distinguishing brain activity from artefacts.
- **R5** ...seeing how a component's activity is distributed across the scalp sensors.
- **R6** ...logging and exporting classification decisions for reproducibility and team consistency.

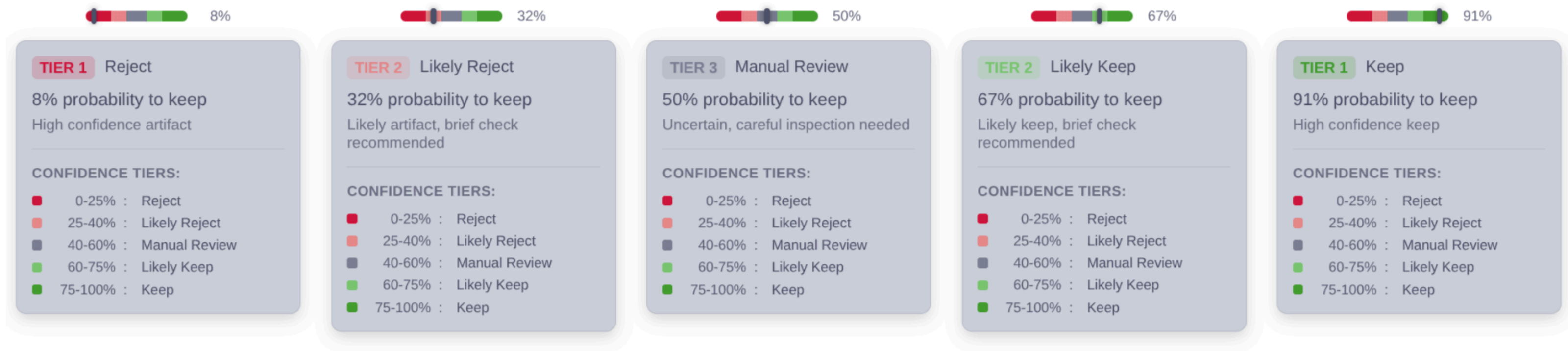
System Architecture



Classification Workflow

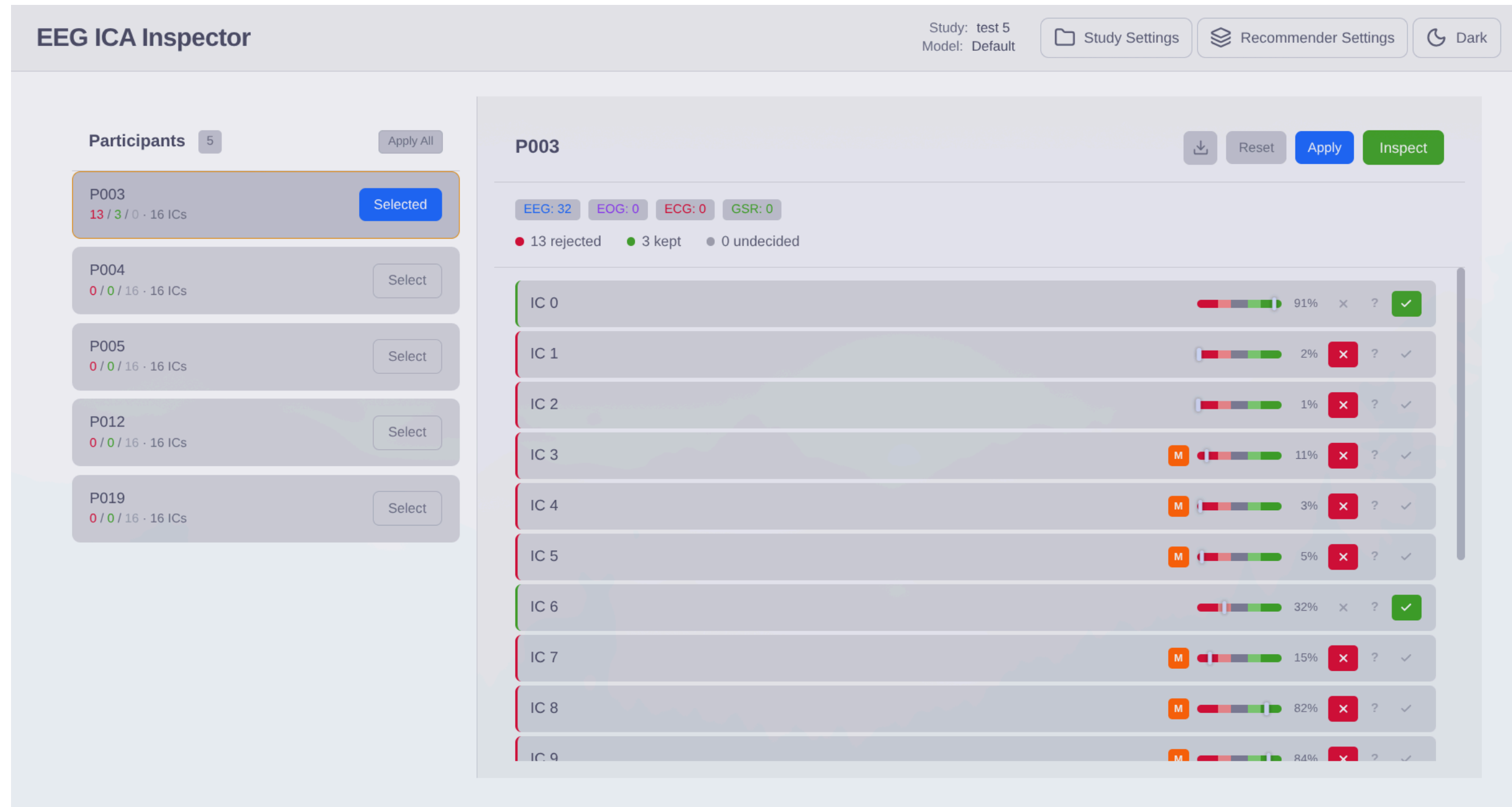


Confidence Recommender

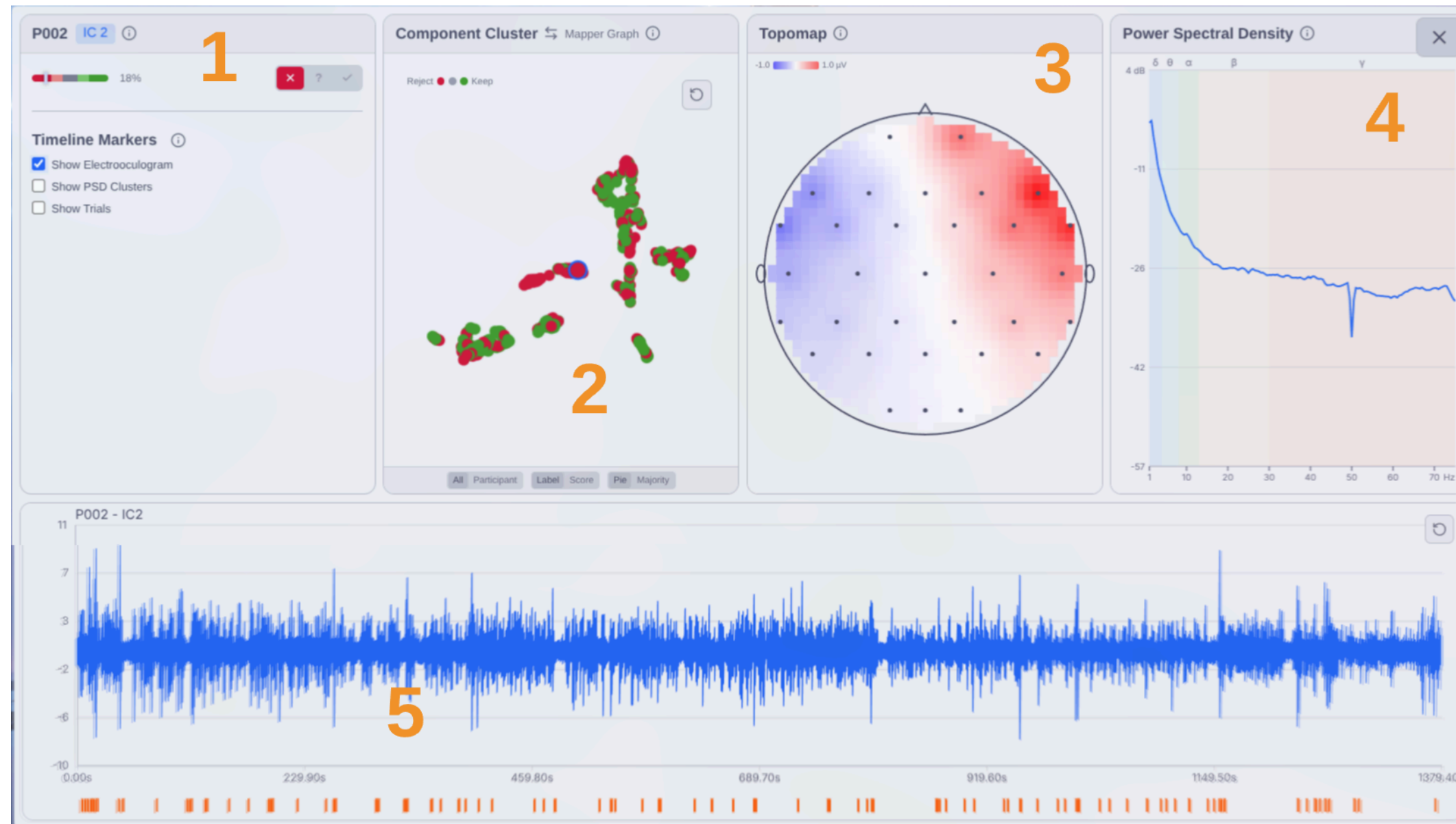


- Tier 1: high-confidence
- Tier 2: quick review
- Tier 3: manual inspection

Participants and Components Overview



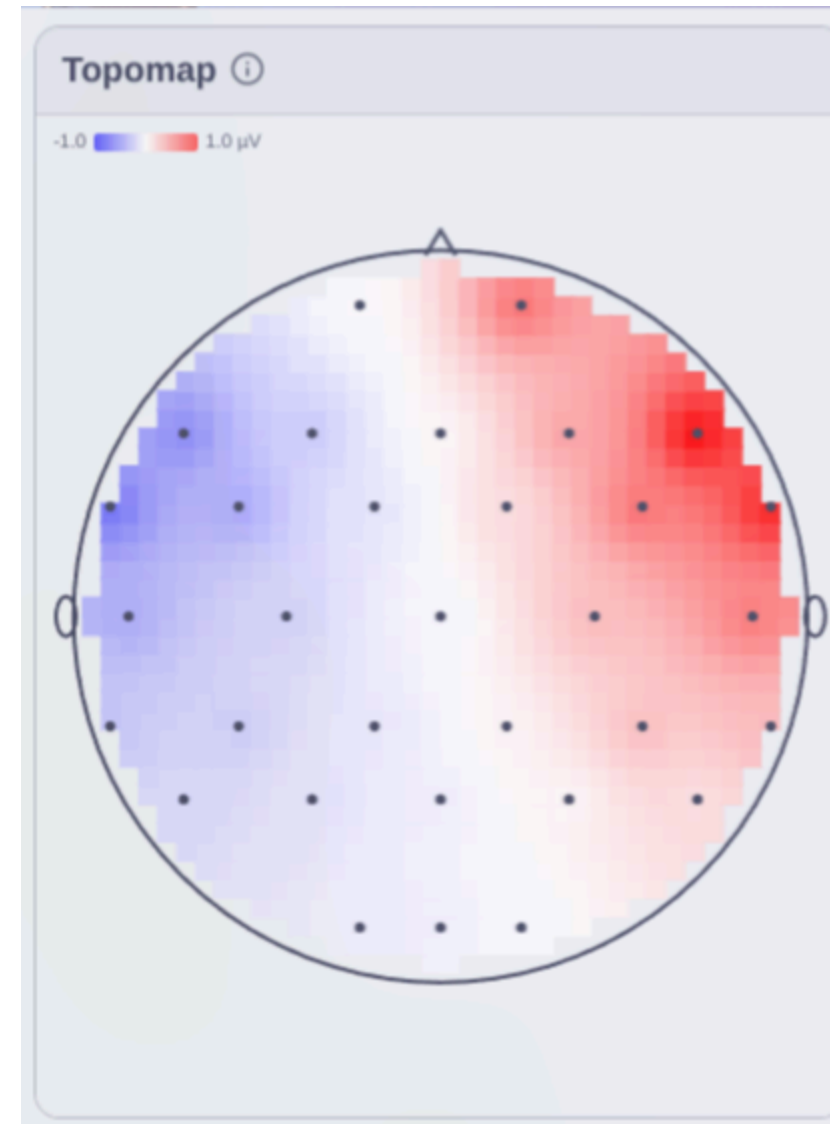
Component Inspector



Component Cluster and Mapper Graph



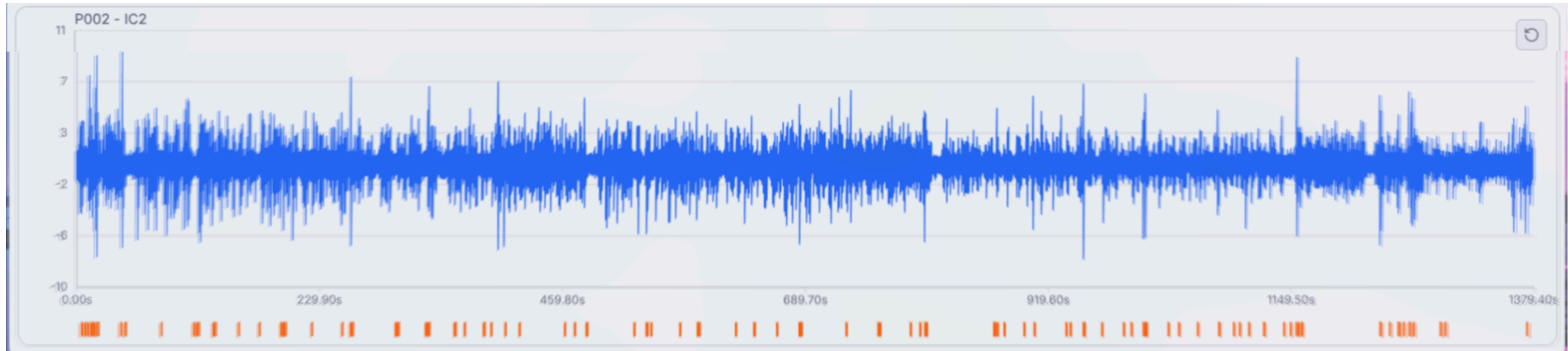
Topomap



Power Spectral Density



Timeline



Overlays: EOG markers · trial markers · PSD clusters

PSD Clustering



K-means on windowed spectra colour-codes the timeline where a component's frequency profile changes

Evaluation

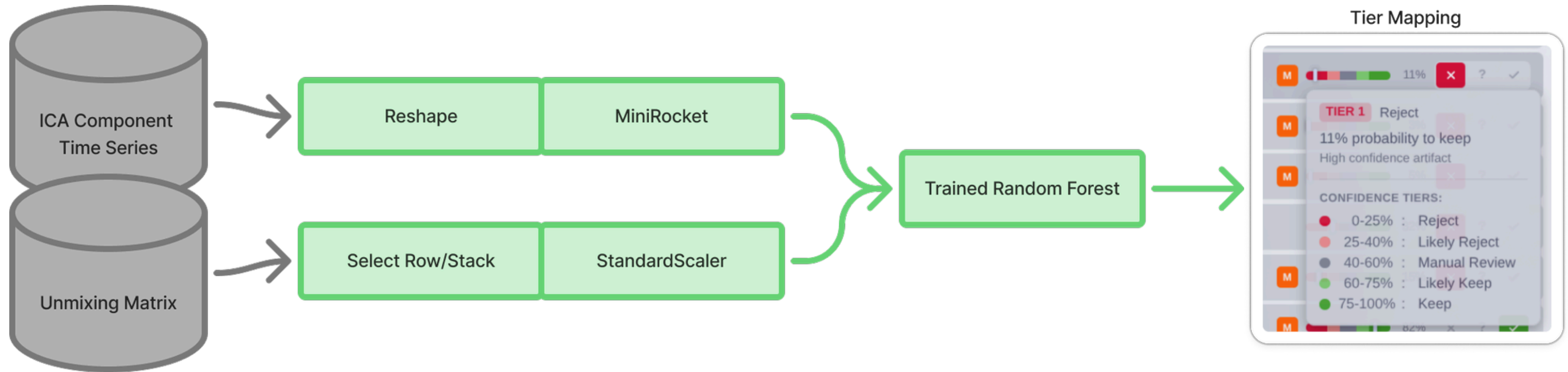
1. **Recommendations:** review the Tier 1 recommendations, accept three correct ones, and find at least one to overrule
2. **Artefact identification:** find a component that could be an eye blink and validate it with the topographic map
3. **Band power analysis:** examine a low-scored component's band-power clusters on the timeline and interpret the segments
4. **Mapper exploration:** using the Mapper Graph, find a mostly-reject cluster, name its artefact type, and re-check any keep labels
5. **View comparison:** compare the Component Cluster and the Mapper Graph and discuss which is more helpful



Backup

Q&A support. Press O for overview.

Recommender Pipeline



- MiniRocket features + 16 topography weights → keep-probability → tiers
- 480 components / 30 participants · ~75% test accuracy

Classification Experiments

Exp	Features	CV	Test	Rej. rec.
01	MiniRocket + Ridge (single split)	—	77%	58%
02	MiniRocket + Ridge + CV	70%	70%	58%
03	MultiRocket + Ridge + CV	56%	48%	52%
04	MultiRocket + Topography + RF	63%	75%	70%
05	Matrix Profile + Topography + RF (best)	67%	79%	76%
06	MiniRocket + Topography + RF (deployed)	65%	75%	70%
07	All-to-all MP similarity + Topo + RF	67%	72%	58%
08	Multi-scale MP + Spectral + Topo (6 models)	65–73%	61–69%	—
09a	Feature selection (RFE, 20 of 40)	—	70%	—

Data Flow

